

AGRISMART AI

Crop Disease Detection - Model Report

1. TASK

Task Type: Crop-disease image classification across 38 distinct classes.

Input & Output: Input: RGB leaf images (224x224 resolution). Output: Multi-class probability distribution identifying specific plant disease or healthy state.

2. DATASET & SPLIT

Dataset Source: PlantVillage color dataset (54,305 total images across 38 classes). Link: [PlantVillage Dataset](#)

Development Split: Split Ratio: 80% Train / 10% Validation / 10% Test (Random Seed: 42). Exact Sizes: Train = 43,429 | Validation = 5,417 | Test = 5,459 images.

Note: This development split is distinct from the official SIH held-out field-test split.

3. MODEL/APPROACH

Architecture: EfficientNet-B0 (ImageNet pretrained), Input size: 224x224 RGB. Classifier: Dropout (0.3) + Linear layer (38 outputs). Best checkpoint selected via validation Macro-F1.

Hyperparameters: Loss: CrossEntropyLoss | Optimizer: AdamW (LR=0.0001, Weight Decay=0.0001) | Batch Size: 32 | Epochs: 5 | Scheduler: ReduceLROnPlateau (factor=0.5, patience=2).

Augmentation: Random horizontal flip, random rotation +15°, ColorJitter, ImageNet normalization.

4. METRIC & RESULT

Current Development Test Performance: Macro-F1 (Primary) = 99.47% | Accuracy = 99.63%

Detailed Test Metrics: Macro Precision = 99.52% | Macro Recall = 99.43% | Weighted F1 = 99.63% | Test Loss = 0.0114

Sample Count: Total Test Samples = 5,459 | Correct Predictions = 5,439 | Incorrect Predictions = 20

Confusion Matrix Analysis:

- Only 20 out of 5,459 test images were incorrectly classified.
- Largest confusion: Corn Cercospora / Gray Leaf Spot → Northern Leaf Blight (6 instances).
- Reverse confusion: Corn Northern Leaf Blight → Cercospora / Gray Leaf Spot (2 instances).
- Tomato confusion: Tomato Late Blight → Early Blight (3 instances).

Class	P	R	Class	P	R	Class	P	R
Apple Apple scab	100.00%	100.00%	Grape Leaf blight	100.00%	100.00%	Strawberry - Leaf scorch	100.00%	100.00%
Apple Black rot	100.00%	100.00%	Grape healthy	100.00%	100.00%	Strawberry - healthy	100.00%	100.00%

Class	P	R	Class	P	R	Class	P	R
Apple Cedar apple rust	100.00%	100.00%	Orange Huanglongbing	100.00%	100.00%	Tomato - Bacterial spot	100.00%	99.53%
Apple - healthy	100.00%	100.00%	Peach Bacterial spot	100.00%	100.00%	Tomato Early blight	96.15%	100.00%
Blueberry healthy	99.34%	100.00%	Peach-healthy	100.00%	100.00%	Tomato - Late blight	98.95%	97.92%
Cherry Powdery mildew	100.00%	99.06%	Pepper - Bacterial spot	100.00%	100.00%	Tomato Leaf Mold	98.96%	98.96%
Cherry - healthy	100.00%	98.84%	Pepper-healthy	99.33%	100.00%	Tomato Septoria leaf spot	99.44%	100.00%
Corn Cercospora/ Gray leaf spot	95.83%	88.46%	Potato Early blight	100.00%	100.00%	Tomato Spider mites	100.00%	100.00%
Corn Common rust	100.00%	100.00%	Potato Late blight	100.00%	100.00%	Tomato - Target Spot	100.00%	97.87%
Corn Northern Leaf Blight	94.17%	97.98%	Potato healthy	100.00%	100.00%	Tomato Tomato Yellow Leaf Curl Virus	99.81%	99.81%
Corn - healthy	100.00%	100.00%	Raspberry healthy	100.00%	100.00%	Tomato Tomato mosaic virus	100.00%	100.00%
Grape Black rot	100.00%	100.00%	Soybean - healthy	99.80%	100.00%	Tomato healthy	100.00%	100.00%
Grape - Esca	100.00%	100.00%	Squash Powdery mildew	100.00%	100.00%			

5. LIMITATIONS

- PlantVillage images are largely clean and lab-like compared with real-world field photographs.
- Real field images may present complex backgrounds, lighting variations, blur, occlusion, and varying disease severity stages.
- The primary failure modes occur between visually similar diseases (notably within corn and tomato classes).
- The current 99.47% result reflects the development test split and must NOT be presented as the official SIH field-test score if an unseen field-condition test set is required.

6. WORKING MODEL

For the complete working model and implementation, please refer to the Google Colab notebook linked below:-

<https://colab.research.google.com/drive/1WqBLSFT44HKN8Ui6zH3Ft02SnOAYEJ9D?usp=sharing>